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Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 1

- 1. Title:** Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.
- 3. Theory:** JavaScript is a scripting language used to make web pages interactive. It can be added in three ways:
 - Inline JavaScript: Written inside HTML elements.
 - Internal JavaScript: Written inside the `<script>` tag in the HTML file.
 - External JavaScript: Written in a separate `.js` file and linked to the HTML page.
 - Console methods like `console.log()`, `console.info()`, `console.warn()`, and `console.error()` are used for debugging and displaying messages.



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3. Output:

The screenshot displays a web browser window titled 'Shreya Maske - JS Exp 1' with the address bar showing the file path. The webpage content includes:

- JavaScript Experiment 1**
- Inline JavaScript (Welcome)**: A button that triggers a message: 'WELCOME to JavaScript Lab Shreya!'
- Internal JavaScript (Info)**: A button that triggers a display of user information: 'Name: Shreya Maske', 'PRN: 24070521196', 'Branch: CSE', and 'Year: Third Year'.
- External JavaScript**: A note to 'Open Console to see console method outputs'.

The Chrome DevTools console on the right shows the following:

- Log for 'Output JavaScript Experiment 1' at js1.js:3.
- Log for 'Information about Shreya Maske' at js1.js:5.
- Warning message: 'This is a warning message.' at js1.js:7.
- Error message: 'This is an error message.' at js1.js:9.
- Log for an array: `Array(1) [0: {Name: 'Shreya Maske', PRN: '24070521196', Branch: 'CSE', Year: 'Third Year'}, length: 1]` at js1.js:11.
- Uncaught exception: 'Unsafely attempt to load URL' at webpage.html:1, with a message about file:// URLs being treated as unique security origins.

4. Result/Conclusion: The experiment was performed successfully. A webpage displaying a welcome message and user information was created using inline, internal, and external JavaScript, and console methods were demonstrated successfully.



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Case Study No.: 1

1. **Case Study Title:** Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
!DOCTYPE html>
<html>
  <head>
    <title>Shreya 24070521196</title>
    <!-- External JavaScript -->
    <script src="script.js"></script>
  </head>
  <body style="text-align:center;background-color: lightblue;">

    <h1>Symbiosis Institute of Technology, Nagpur</h1>
    <h2>Department of Computer Science and Engineering</h2>
    <hr style="border:2px solid red;"><hr style="border:2px solid red;">
    <h3 style="font-size:30px;">Student Information</h3>
    <a href="shinfo.html" target="_blank" style=font-size:30px;>Go to Student
Information Page</a>

    <br><br><br><br>
    <!-- Inline JavaScript -->
    <button onclick="alert('Welcome to SIT Nagpur')"style=font-size:30px;>
      alert
    </button>
    <br><br><br><br>
    <p style="font-size:30px;"><b>Open Console to see console.log()
output.</b></p>
  </body>
</html>
```



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```
<!DOCTYPE html>
<html>
<head>
  <title>Shreya Info 24070521196</title>
</head>
<body>
  <h1>Student Information</h1>
  <p><b>Name:</b> Shreya Maske</p>
  <p><b>PRN:</b> 24070521196</p>
  <p><b>Course:</b> B.Tech</p>
  <p><b>Branch:</b> Computer Science and Engineering</p>
  <p><b>Year:</b> Third Year</p>
  <br>
</body>
</html>
```

```
function welcomeExternal() {
  alert("Welcome to SIT Nagpur");
}

console.log("Welcome to SIT Nagpur");
```



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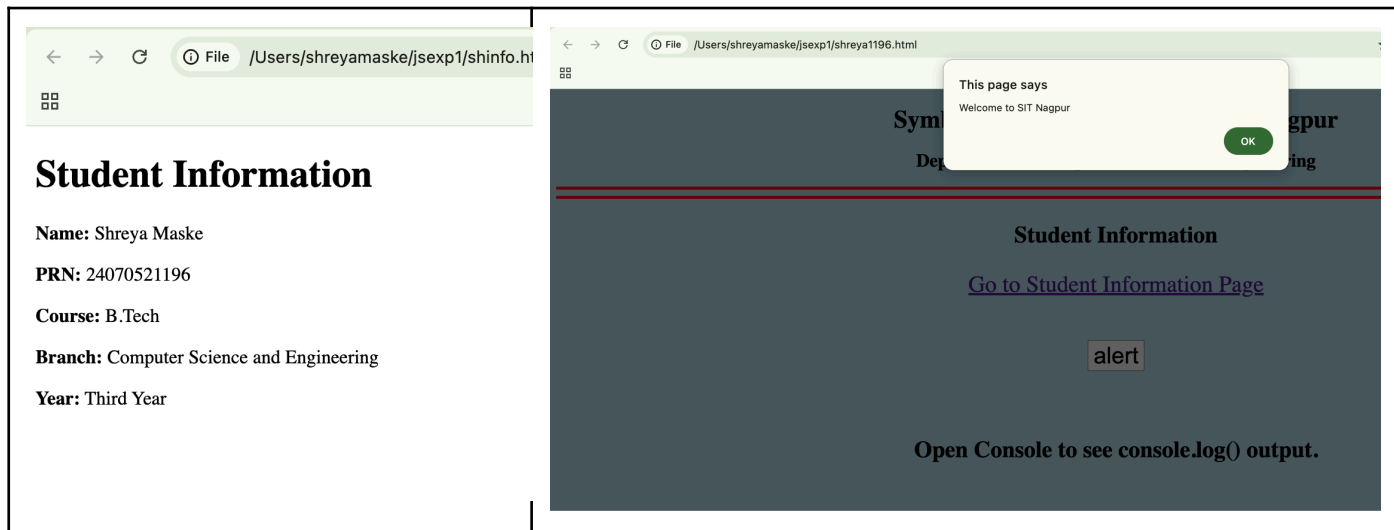
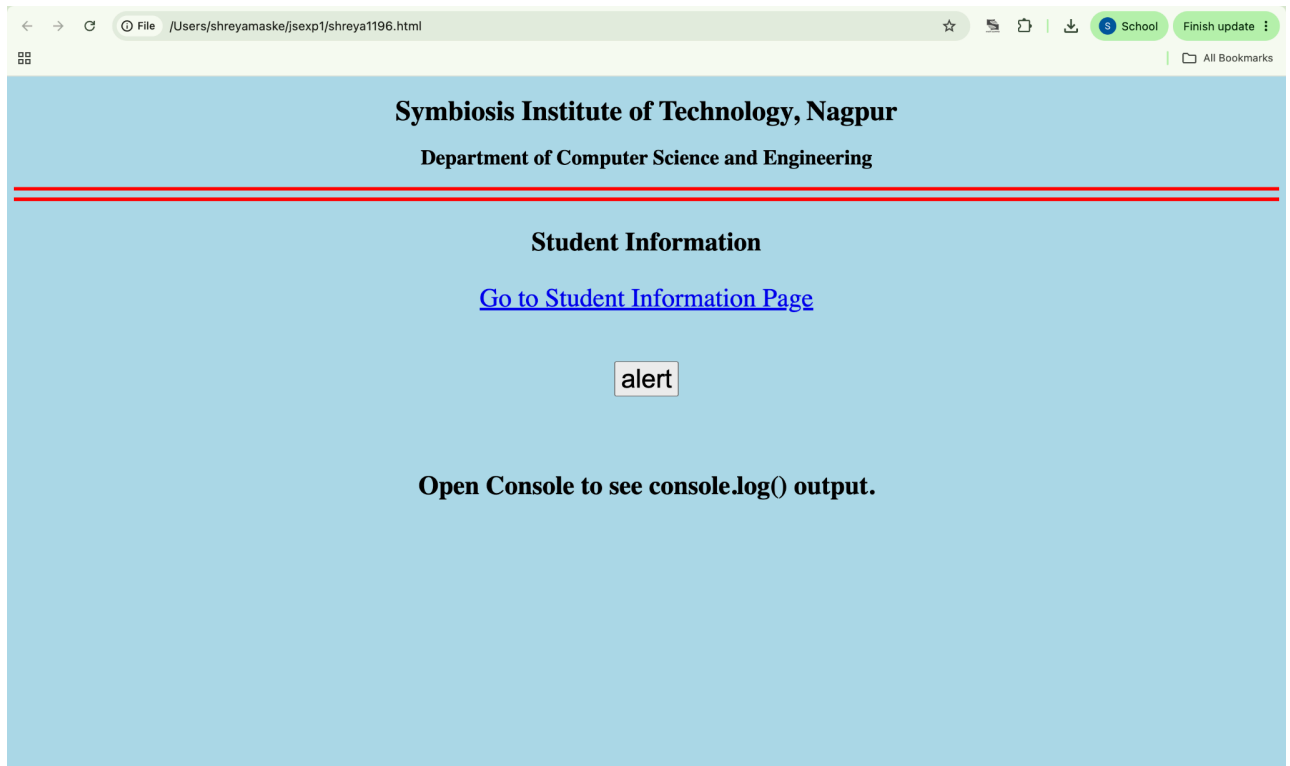
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3. Output:



3. **Result/Conclusion:** The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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● Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$${}`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :

The image shows two screenshots of a web browser displaying a 'Billing Calculator' application. The browser tabs include 'sec_appti_24-28: Attendance' and 'Billing Calculator'. The address bar shows the file path: '/Users/pratham/jsexp2/shreya1196.html'.

The first screenshot shows a prompt box titled 'This page says' with the text 'Enter Customer Name :'. The input field contains 'Shreya Maske'. There are 'Cancel' and 'OK' buttons.

The second screenshot shows a similar prompt box titled 'This page says' with the text 'Enter Product Name :'. The input field contains 'Microwave'. There are 'Cancel' and 'OK' buttons.



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The screenshot shows a web browser with two tabs: 'sec_apti_24-28: Attendance' and 'Billing Calculator'. The address bar shows the file path '/Users/pratham/jsexp2/shreya1196.html'. A dialog box titled 'This page says' is open, prompting the user to 'Enter Quantity :'. The input field contains the value '1'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the same web browser with the 'Billing Calculator' tab active. A dialog box titled 'This page says' is open, prompting the user to 'Enter Price per Item :'. The input field contains the value '42000'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the 'Shreya's Billing Calculator' web application. The title is 'Shreya's Billing Calculator'. Below the title is a section titled 'Billing Reciept' (note the spelling). The receipt details are as follows:

- Customer: Shreya Maske
- Product: Microwave
- Quantity: 1
- Price per item: 42000
- Total: ₹42000
- GST (18%): ₹7560.00
- Final Bill: ₹49560.00**

5. **Result/Conclusion:** A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

1. **Case Study Title:** Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

2. Case Study Program Code:

```
shreya1196.html case2.html X
case2.html > html > body > script > bill
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>Online Shopping Bill</title>
5 </head>
6 <body style="background-color: ■ rgb(230, 218, 173);">
7 <h1>Shreya's Online Shopping Website</h1>
8 <script>
9 const GST = 0.18;
10 // Customer Details
11 var custname = prompt("Enter Customer Name:");
12 var mobile = prompt("Enter Mobile Number:");
13 var invoice = prompt("Enter Invoice Number:");
14 var product = prompt("Enter Product Name:");
15 var quantity = Number(prompt("Enter Quantity (Kg):"));
16 var rate = Number(prompt("Enter Rate per Kg:"));
17 var discount = Number(prompt("Enter Discount (%):"));
18 var packing = Number(prompt("Enter Packing Charges:"));
19 var payment = prompt("Enter Payment Mode (Cash/Card/UPI):");
20 var membership = prompt("Membership (Yes/No):");
21 // Calculations
22 let total = quantity * rate;
23 let discountAmt = total * discount / 100;
24 let subtotal = total - discountAmt;
25 let gstAmt = subtotal * GST;
26 let finalBill = subtotal + gstAmt + packing;
```



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```
27 // Object
28 const bill = {
29     customer: custname,
30     mobile,
31     invoice,
32     product,
33     quantity,
34     rate,
35     discount,
36     packing,
37     payment,
38     membership,
39     total,
40     discountAmt,
41     gstAmt,
42     finalBill
43 };
44 // Destructuring
45 const {
46     customer,
47     mobile: mob,
48     invoice: inv,
49     product: pname,
50     payment: payMode,
51     membership: member,
52     gstAmt: gst,
53     finalBill: amount
54 } = bill;
55 // Output
56 document.write(`<h2>Shopping Invoice</h2>`);
57 document.write(`<hr>`);
58 document.write(`<p><b>Customer Name:</b> ${customer}</p>`);
59 document.write(`<p><b>Mobile No:</b> ${mob}</p>`);
60 document.write(`<p><b>Invoice No:</b> ${inv}</p>`);
61 document.write(`<p><b>Product:</b> ${pname}</p>`);
62 document.write(`<p><b>Quantity:</b> ${quantity} Kg</p>`);
63 document.write(`<p><b>Rate per Kg:</b> ₹${rate}</p>`);
64 document.write(`<p><b>Total:</b> ₹${total.toFixed(2)}</p>`);
65 document.write(`<p><b>Discount:</b> ₹${discountAmt.toFixed(2)}</p>`);
66 document.write(`<p><b>GST (18%):</b> ₹${gst.toFixed(2)}</p>`);
67 document.write(`<p><b>Packing Charges:</b> ₹${packing.toFixed(2)}</p>`);
68 document.write(`<p><b>Payment Mode:</b> ${payMode}</p>`);
69 document.write(`<p><b>Membership:</b> ${member}</p>`);
70 document.write(`<h2>Final Bill: ₹${amount.toFixed(2)}</h2>`);
71 </script>
72 </body>
73 </html>
```



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3. Output:

← → ↺ ⓘ File /Users/shreyamaske/jsexp2/case2.html

☐☐

Shreya's Online Shopping Website

Shopping Invoice

Customer Name: Shreya Maske

Mobile No: 7798642334

Invoice No: 880024

Product: Basmati Rice

Quantity: 6 Kg

Rate per Kg: ₹120

Total: ₹720.00

Discount: ₹14.40

GST (18%): ₹127.01

Packing Charges: ₹35.00

Payment Mode: Card

Membership: No

Final Bill: ₹867.61

4. Result/Conclusion: The online shopping billing webpage was successfully developed using **HTML, CSS, and JavaScript**. The program accepted customer and product details, calculated the total bill including **discount, GST, and packing charges**, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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● Experiment No.: 3

1. **Title:** Implement control structures and form validation; create a grading system based on user-entered marks.

2. **Software/Tools Required:** VS Code, web browser, HTML, JavaScript.

3. **Theory: Control structures** (`if`, `else if`, and `else`) are used to make decisions based on the marks entered by the user.

Form validation checks whether the input is empty or outside the valid range (0–100) before processing.

The `getElementById()` method is used to access the value entered in the input box.

The `Number()` function converts the entered value from text to a numeric value. A **function** (`calculateGrade()`) is called when the **Submit** button is clicked to validate the input and calculate the grade.

The calculated grade is displayed on the webpage using the `innerHTML` property.

4. Output :

The screenshot shows a web browser window with the address bar displaying the file path: `/Users/shreyamaske/jsexp3/Shreya1196exp3.html`. The page has a purple background and features the title "Shreya's 1196 Grading System" in a large, bold, black font. Below the title, the text "Enter Marks:" is displayed. Underneath, there is a text input field with the placeholder text "Enter Marks here". Below the input field is a button labeled "Submit".



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The screenshot shows a web browser window titled "Grading System" with a file path in the address bar: "/Users/shreyamaske/jsexp3/Shreya1196exp3.html". The page has a dark purple background and displays the title "Shreya's 1196 Grading System". Below the title, it says "Enter Marks:" followed by a text input field containing the number "106". A "Submit" button is located below the input field. A white error message box is overlaid on the right side of the page, stating "This page says Enter Valid Marks between 0 and 100" with an "OK" button.

This screenshot is similar to the previous one, showing the same web application. However, the text input field now contains the placeholder text "Enter Marks here". The error message box is still present, but the message has changed to "This page says Enter Marks" with an "OK" button.

The screenshot shows the final output of the grading system. The page has a light purple background. It displays the title "Shreya's 1196 Grading System". Below the title, it says "Enter Marks:" followed by a text input field containing the number "74". A "Submit" button is located below the input field. Below the input field, the results are displayed: "Marks: 74" and "Grade: B+".

5. Result/Conclusion: The grading system was successfully implemented using JavaScript. The program validated the user input, assigned grades using if-else control structures, and displayed the correct grade based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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Case Study No.: 3

1. **Case Study Title:** Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Write a JavaScript program that accepts student marks and display the Scholarship category.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Scholarship Eligibility</title>
</head>
<body style="background-color:rgba(176, 230, 187, 0.867);">
  <h1>SIT Scholarship Eligibility</h1> <hr><hr>
  <label>Enter Marks:</label><br><br>
  <input type="text" id="marks" placeholder="Enter Marks
here"><br><br>
  <button onclick="calculateEligibility()">Submit</button>
  <h2 id="result"></h2>
  <script>
    function calculateEligibility(){
      let marks=document.getElementById("marks").value;
      if(marks == "")
      {
        alert("Enter Marks");
        return;
      }
    }
  </script>
</body>
</html>
```



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```
marks = Number(marks);  
if(marks < 0 || marks > 100)  
{  
    alert("Enter Valid Marks for Eligibility");  
    return;  
}  
let grade;  
if(marks >= 95)  
    grade = "Platinum Scholarship";  
else if(marks >= 85)  
    grade = "Gold Scholarship";  
else if(marks >= 70)  
    grade = "Silver Scholarship";  
else if(marks >= 50)  
    grade = "Bronze Scholarship";  
else if(marks <= 50)  
    grade = "Not Eligible for Scholarship";  
document.getElementById("result").innerHTML =  
    "Marks: " + marks + "<br>Grade: " + grade;  
}  
</script>  
</body>  
</html>
```



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3. Output:

Screenshot of the SIT Scholarship Eligibility form in a web browser. The browser tab is "Scholarship Eligibility" and the address bar shows the file path. The form has a green header with the title "SIT Scholarship Eligibility". Below the title is a label "Enter Marks:" followed by a text input field containing "Enter Marks here" and a "Submit" button.

Screenshot of the SIT Scholarship Eligibility form with a tooltip. The tooltip says "This page says Enter Marks" and has an "OK" button. The form is the same as the previous screenshot but with a dark grey background.

Screenshot of the SIT Scholarship Eligibility form with the input field filled with "32". Below the form, the text "Marks: 32" and "Grade: Not Eligible for Scholarship" is displayed.



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Scholarship Eligibility

File /Users/shreyamaske/jsexp3/shreya1196case3.html



SIT Scholarship Eligibility

Enter Marks:

Submit

Marks: 86
Grade: Gold Scholarship

4. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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● Experiment No.: 4

1. **Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. **Software/Tools Required:** VS Code, JavaScript.

3. Theory:

Anonymous Function: A function without a name, assigned to a variable and used to display the welcome message.

Global Scope: Variables declared outside functions (e.g., language) are accessible throughout the program.

Local Scope: Variables declared inside a function (e.g., topic) are accessible only within that function.

Closure: A function that retains access to its outer function's variables even after the outer function has finished execution. The counter() function demonstrates this by preserving the count variable.

Arrow Function: A concise syntax for writing functions, used here to display output messages.

Try-Catch: Used for exception handling to catch and handle invalid input without stopping program execution.

Palindrome Checker: Compares the original string with its reversed version. If both are equal, the string is a palindrome; otherwise, it is not.

4. Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
1221 is a palindrome.
```



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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
abc1221cba is a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
daman is not a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
Error: Input must be a string.
```

5. Result/Conclusion: The program successfully demonstrates the use of JavaScript function types, variable scope, closures, arrow functions, and exception handling using try-catch. It correctly checks whether a given string is a palindrome while illustrating important JavaScript programming concepts in a simple and effective manner.



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Case Study No.: 4

- Case Study Title:** Real time case study ATM card pin verification using JavaScript functions case study an ATM system verifies whether a customers pin is a palindrome. If the pin reads the same forward and backward the Bank displays a special security message. To implement this, create a function to reverse the entered pin number and another function to check whether it is a palindrome. use function declaration, function expression, Arrow function and demonstrate scope and closure concepts.

Case Study Program Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>ATM PIN Verification</title>

</head>

<body style="background-color: rgb(233, 166, 58);">

<h2 >Shreya's (1196) ATM PIN Verification System</h2>

Enter ATM PIN:

<input type="text" id="pin">

<button onclick="verifyPIN()">Verify PIN</button>

<hr>

<div id="output"></div>

<script src="shreya1196case4.js"></script>

</body>

</html>
```



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```
// Anonymous Function (Function Expression)
let welcome = function () {
    return "<h3>ATM PIN Verification System</h3>";
};

// Global Variable
let bankName = "Shreya's (1196) Bank";

// Function Declaration - Scope
function scopeDemo() {
    let branch = "Nagpur"; // Local Variable
    return "<b>Bank:</b> " + bankName +
        "<br><b>Branch:</b> " + branch + "<br>";
}

// Closure
function counter() {
    let count = 0;
    return function () {
        count++;
        return count;
    };
}

let c = counter();

// Function Declaration - Reverse PIN
function reversePIN(pin) {
    return pin.split("").reverse().join("");
}

// Function Declaration - Check Palindrome
function palindrome(pin) {
    let reverse = reversePIN(pin);
    return pin === reverse;
}
```



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```
// Arrow Function
const display = (message) => {
    document.getElementById("output").innerHTML +=
message + "<br>";
};
// Main Function
function verifyPIN() {
    document.getElementById("output").innerHTML =
"";
    display(welcome());
    display(scopeDemo());
    display("<b>Verification Attempt:</b> " + c());
    try {
        let pin =
document.getElementById("pin").value;
        if (pin === "") {
            throw "Please enter ATM PIN. ";
        }
        if (palindrome(pin)) {
            display("<h3 style='color:red'>Weak
Security!</h3>");
            display("PIN is a palindrome.");
            display("Choose a stronger PIN.");
        }
        else {
            display("<h3 style='color:green'>Strong
Security!</h3>");
            display("PIN is not a palindrome.");
        }
    }
    catch(error) {
        display("<b>Error:</b> " + error);
    }
}
```



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}

3. Output:

24070521196_Shreya Maske_ x

ATM PIN Verification x +

< > ↺

File /Users/shreyamaske/jsexp4/shreya1196case4.html

⌵

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN: Verify PIN

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 1

Strong Security!

PIN is not a palindrome.



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24070521196_Shreya Maske_ x

ATM PIN Verification x +

< > ↺

File /Users/shreyamaske/jsexp4/shreya1196case4.html

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN:

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 2

Weak Security!

PIN is a palindrome.
Choose a stronger PIN.

4. Result/Conclusion:

The ATM PIN verification system was successfully implemented using JavaScript. The program correctly identified whether the entered PIN was a palindrome, displayed "**Weak Security**" for palindrome PINs and "**Strong Security**" for non-palindrome PINs, while demonstrating function declaration, function expression, arrow function, scope, closures, and exception handling using try-catch.



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● Experiment No.: 5

1. **Title:** Apply array methods and object handling, create a cart total calculator with discount logic.

2. **Software/Tools Required:** VS Code, JavaScript.

3. Theory:

- 1) The program uses an array to store multiple shopping cart items.
- 2) Each item is represented as an object containing name, price, and quantity.
- 3) User input is accepted using `prompt()` and converted into numeric values using `parseInt()` and `parseFloat()`.
- 4) The `for` loop is used to read details of multiple items.
- 5) The `push()` method stores each item object in the cart array.
- 6) The `reduce()` method calculates the total amount by summing price $\times$ quantity of all items.
- 7) The discount percentage is calculated based on the total purchase amount using `Math.floor()`.
- 8) An `if` statement is used to limit the maximum discount and apply the appropriate discount.
- 9) discount based on the total purchase amount, where every ₹1000 increase in the total bill increases the discount by 10%.
- 10) The final payable amount is calculated by subtracting the discount from the total amount.



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4. Output :

```
pratham@Prathams-MacBook-Air-2 jsexp5 % node "/Users/shreyamaske/jsexp5/shreya_1196_prac5.js"
Enter number of items:3
Enter item name:Flowerpot
Enter price:750
Enter quantity:2
Enter item name:Mat
Enter price:350
Enter quantity:1
Enter item name:Frame
Enter price:500
Enter quantity:1
=====
SHREYA'S (1196) CART CALCULATOR
=====
Total: 2350
Discount: 470
Final Amount: 1880
```

5. Result/Conclusion:

The shopping cart total calculator was successfully implemented using JavaScript. The program accepted user input, stored item details using arrays and objects, calculated the total amount using the `reduce()` method, applied the appropriate discount based on the purchase amount, and displayed the final payable amount correctly.



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Case Study No.: 5

1. Case Study Title: Implementation of JavaScript Array Operations Using Built-in Methods

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Array Operations</title>
  <style>
    body {
      background-color: lightblue;
    }
  </style>
</head>
<body>
<h2>JavaScript Array Operations</h2>
<label>Enter array elements:</label>
<input type="text" id="arrayInput">
<br><br>
<label>Value for Push:</label>
<input type="number" id="pushValue">
<br><br>
<label>Value for Unshift:</label>
<input type="number" id="unshiftValue">
<br><br>
<label>Splice Index:</label>
<input type="number" id="spliceIndex">
<label>Splice Value:</label>
<input type="number" id="spliceValue">
<br><br>
<label>Slice Start:</label>
```



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```
<input type="number" id="sliceStart">

<label>Slice End:</label>
<input type="number" id="sliceEnd">
<br><br>
<button onclick="arrayOperations()">Perform
Operations</button>
<p id="output"></p>
<script src="shreya_1196_case5.js"></script>
</body>
</html>
```

```
function arrayOperations() {
    let arr = document.getElementById("arrayInput").value
        .split(", ").map(Number);
    let pushValue =
Number(document.getElementById("pushValue").value);
    let unshiftValue =
Number(document.getElementById("unshiftValue").value);

    let spliceIndex =
Number(document.getElementById("spliceIndex").value);
    let spliceValue =
Number(document.getElementById("spliceValue").value);

    let sliceStart =
Number(document.getElementById("sliceStart").value);
    let sliceEnd =
Number(document.getElementById("sliceEnd").value);
    let output = "";
    output += "Original Array: " + arr + "<br><br>";
    arr.push(pushValue);
    output += "After push: " + arr + "<br>";
```



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```
// pop()
arr.pop();
output += "After pop: " + arr + "<br>";

// unshift()
arr.unshift(unshiftValue);
output += "After unshift: " + arr + "<br>";

// shift()
arr.shift();
output += "After shift: " + arr + "<br>";

// splice()
arr.splice(spliceIndex, 1, spliceValue);
output += "After splice: " + arr + "<br>";

// slice()
let newArray = arr.slice(sliceStart, sliceEnd);
output += "After slice: " + newArray + "<br><br>";

// Maximum and Minimum
let maximum = Math.max(...arr);
let minimum = Math.min(...arr);

output += "Maximum: " + maximum + "<br>";
output += "Minimum: " + minimum;

document.getElementById("output").innerHTML = output;
}
```



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3. Output:

← → 🔄 📄 File /Users/shreyamaske/javascript_exp/jsexp5/shreya_1196_case5.html

☐☐☐

JavaScript Array Operations

Enter array elements:

Value for Push:

Value for Unshift:

Splice Index: Splice Value:

Slice Start: Slice End:

Original Array: 2,34,56,77,81,45,11

After push: 2,34,56,77,81,45,11,90

After pop: 2,34,56,77,81,45,11

After unshift: 1,2,34,56,77,81,45,11

After shift: 2,34,56,77,81,45,11

After splice: 2,34,56,66,81,45,11

After slice: 66,81,45

Maximum: 81

Minimum: 2

4. Result/Conclusion:

Thus, various JavaScript array operations were successfully performed and the results were displayed using user input.



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● Experiment No.: 6

1. Title: Use string functions and regex for email validation, data extraction, and text analysis.

2. Software/Tools Required: VS Code, JavaScript.

3. Theory:

- 1) String functions in JavaScript are used to manipulate and analyze text.
- 2) Regular Expressions (Regex) are used for pattern matching, such as validating and extracting email addresses.
- 3) test() checks whether an email matches the specified email pattern.
- 4) match() extracts email addresses from the given text.
- 5) toUpperCase() and toLowerCase() convert text into uppercase and lowercase.
- 6) length finds the total number of characters in the text.
- 7) includes() checks whether the text contains a specified word such as "Welcome".
- 8) split() divides the text into individual words.
- 9) match(/email/gi) finds the occurrences of the word "email", ignoring case.
- 10) The program demonstrates email validation, data extraction, and text analysis using JavaScript string functions and Regex.



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4. Output :

```
pratham@Prathams-MacBook-Air-2 javascript_exp % node "/Users/shreyamaske/javascript_exp/jsexp6/shreya_1196_prac6.js"
Valid email: true
Extracted emails: [
  'shreyamaske15@gmail.com',
  'shreya.maske.batch2024@sitnagpur.siu.edu.in'
]
Uppercase: CONTACT US AT SHREYAMASKE15@GMAIL.COM OR SHREYA.MASKE.BATCH2024@SITNAGPUR.SIU.EDU.IN . WELCOME TO OUR WEBSITE!
Lowercase: contact us at shreyamaske15@gmail.com or shreya.maske.batch2024@sitnagpur.siu.edu.in . welcome to our website!
Length of text: 110
Contains 'Welcome': true
Words in text: [
  'Contact',
  'us',
  'at',
  'shreyamaske15@gmail.com',
  'or',
  'shreya.maske.batch2024@sitnagpur.siu.edu.in',
  '.',
  'Welcome',
  'to',
  'our',
  'website!'
]
Count of 'email' in text: 0
pratham@Prathams-MacBook-Air-2 javascript_exp %
```

5. Result/Conclusion:

The program successfully implemented string functions and Regular Expressions to validate an email, extract email addresses, perform case conversion, analyze text length and words, check for specific text, and count occurrences of a word.



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Case Study No.: 6

- Case Study Title:** To demonstrate JavaScript string operations using `substring()`, `indexOf()`, `split()`, `replace()`, string reversal, and vowel counting based on user input.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>String Operations</title>
  <style>
    body {
      background-color: rgb(239, 171, 243);
      font-family: Arial;
    }
    input {
      padding: 8px;
      margin: 5px;
      width: 350px;
    }
    textarea {
      padding: 10px;
      margin: 5px;
      width: 500px;
      height: 80px;
    }
    button {
      padding: 10px 20px;
      margin: 10px 5px;
    }
    #output {
      margin-top: 20px;
      font-size: 18px;
    }
  </style>
```



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```
</head>
<body>
  <h2>Shreya's JavaScript String Operations</h2>
  <label>Enter String:</label><br>
  <textarea id="str" placeholder="Enter your string or paragraph"></textarea>
  <br>
  <label>Substring Start:</label>
  <input type="number" id="start" placeholder="Enter start index">
  <br>
  <label>Substring End:</label>
  <input type="number" id="end" placeholder="Enter end index">
  <br>
  <label>IndexOf - Enter text to search:</label>
  <input type="text" id="search" placeholder="Enter character or word">
  <br>
  <label>Split - Enter separator:</label>
  <input type="text" id="separator" placeholder="Example: space">
  <br>
  <label>Replace - Text to replace:</label>
  <input type="text" id="oldText" placeholder="Enter text to replace">
  <br>
  <label>Replace - New text:</label>
  <input type="text" id="newText" placeholder="Enter new text">
  <br>
  <button onclick="performOperations()">Perform Operations</button>
  <div id="output"></div>
  <script src="shreya_case6.js"></script>
</body>
</html>

function performOperations() {

  let str = document.getElementById("str").value;

  let start = Number(document.getElementById("start").value);
  let end = Number(document.getElementById("end").value);

  let search = document.getElementById("search").value;
  let separator = document.getElementById("separator").value;
```



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```
let oldText = document.getElementById("oldText").value;
let newText = document.getElementById("newText").value;
let output = "";
// Substring
output += "Substring: " + str.substring(start, end) + "<br>";
// IndexOf
output += "Index of '" + search + "': " + str.indexOf(search) +
"<br>";
// Split
if (separator === "space") {
    output += "Split: " + str.split(" ") + "<br>";
} else {
    output += "Split: " + str.split(separator) + "<br>";
}
// Replace
output += "Replace: " + str.replace(oldText, newText) + "<br>";

// Reverse
let reverse = str.split("").reverse().join("");
output += "Reversed String: " + reverse + "<br>";
// Count Vowels
let count = 0;
for (let i = 0; i < str.length; i++) {
    let ch = str[i].toLowerCase();
    if (ch == "a" || ch == "e" || ch == "i" ||
        ch == "o" || ch == "u") {
        count++;
    }
}
output += "Number of Vowels: " + count;
document.getElementById("output").innerHTML = output;
}
```



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3. Output:

Fwd: js1ab 7 - shreya.maske.b

String Operations

File /Users/shreyamaske/javascript_exp/jsexp6/shreya_case6.html

Shreya's JavaScript String Operations

Enter String:

My name is Shreya Maske.I love Javascript.
Studying in SIT Nagpur.
24070521196

Substring Start:

Substring End:

IndexOf - Enter text to search:

Split - Enter separator:

Replace - Text to replace:

Replace - New text:

Perform Operations

Substring: me is Shreya Maske.I love Javascript. Studying in SIT Nagpur. 24070521
Index of 'SIT': 55
Split: My name is Shreya Maske,I love Javascript, Studying in SIT Nagpur, 24070521196
Replace: My name is Shreya Maske.I like Javascript. Studying in SIT Nagpur. 24070521196
Reversed String: 69112507042 .rupgaN TIS ni gniydutS .tpircsavaJ evol l.eksaM ayerhS si eman yM
Number of Vowels: 19

4. Result/Conclusion:

Thus, various JavaScript string operations were successfully performed, including **substring extraction**, **text searching**, **splitting**, **replacing**, **reversing**, and **vowel counting**, and the results were displayed dynamically on the webpage.



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● Experiment No.: 7

1. Title: Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

2. Software/Tools Required: VS Code, JavaScript.

3. Theory:

1.DOM (Document Object Model) represents an HTML webpage as a tree of elements that JavaScript can access and modify.

2.DOM Traversal is used to access required HTML elements using methods such as `getElementById()`.

3.DOM Manipulation allows JavaScript to dynamically create, update, and remove webpage elements.

4.`createElement()` creates new elements such as `<li>` and `<button>`.

5.`appendChild()` adds the newly created elements to the task list.

6.`addEventListener()` handles user actions such as button clicks and pressing the Enter key.

7.`classList.toggle()` adds or removes the `completed` class to mark a task as completed.

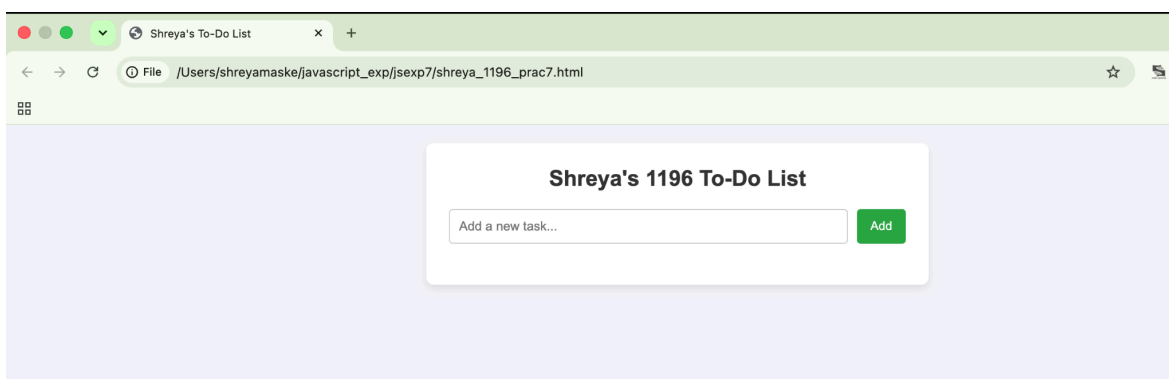
8.`remove()` removes a selected task from the webpage.

9.`trim()` removes extra spaces from the input, while conditional statements prevent empty tasks from being added.

10.`textContent` sets the text content of tasks and buttons.

11.The To-Do List uses these DOM operations to add and manage tasks dynamically without reloading the webpage.

4. Output :





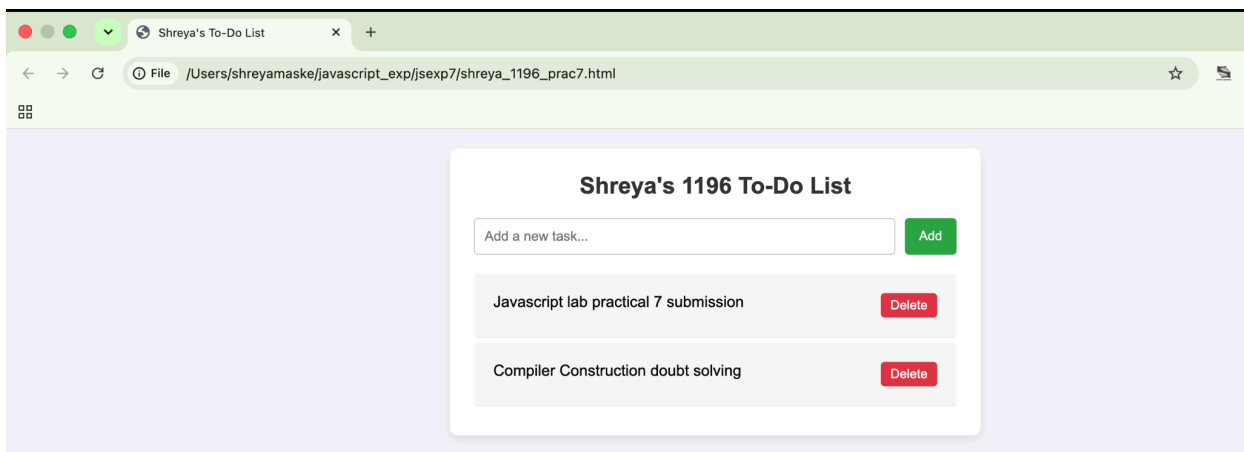
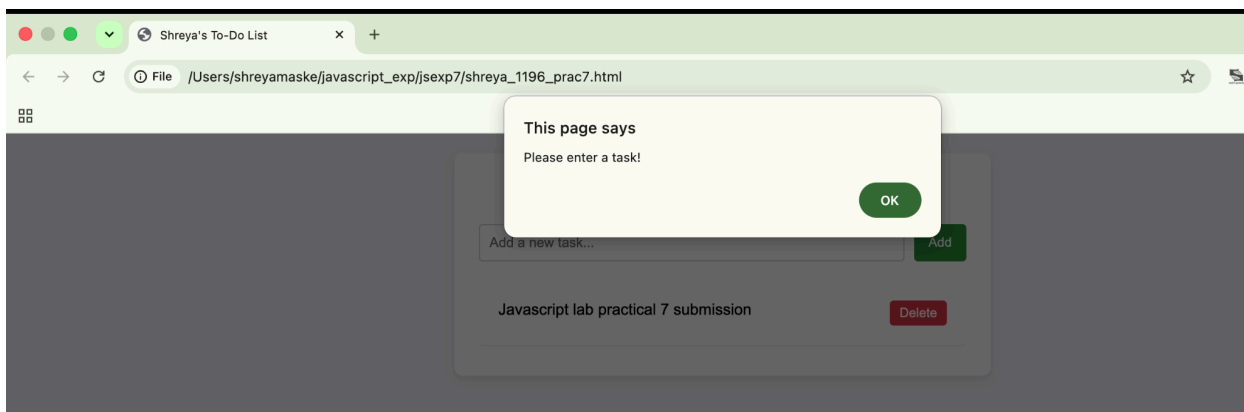
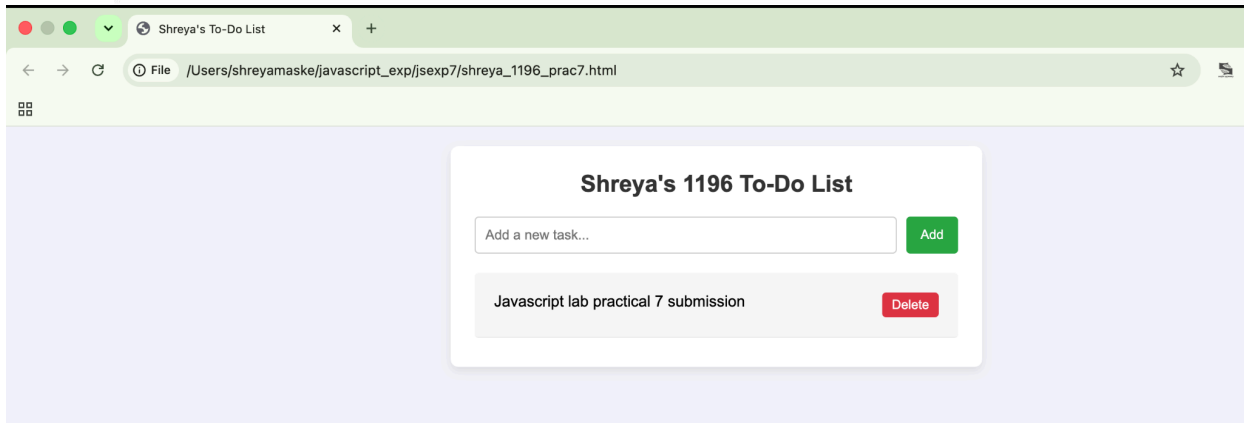
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5. Result/Conclusion:

The To-Do List application was successfully developed using JavaScript DOM manipulation. The program allows users to **add, edit, and delete tasks dynamically**, demonstrating effective use of DOM traversal, DOM updates, and event handling.



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Case Study No.: 7

- 1. Case Study Title:** To demonstrate events handling in javascript, a) access and manipulate from elements using javascript. b) validate user input (eg. email, empty, fields) c) use various form events (submit, change, focus).

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Shreya's (1196) - Event Handling</title>
<style>
body {
font-family: Arial, sans-serif;
background-color: #e8f0fe;
padding: 40px;
}
.form-box {
width: 450px; margin:
auto;
background-color: white; padding: 25px;
border-radius: 10px;
box-shadow: 0 4px 12px gray;
}
h2 {
text-align: center;
}
```



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```
label {
    display: block;
    margin-top: 15px;
    font-weight:
    bold;
}
input, select {
    width: 100%;
    padding: 10px;
    margin-top:
    5px;
    box-sizing: border-box;
}
input:focus,  select:focus
    { border:  2px  solid
    blue; outline: none;
}
.error {
    color: red;
    font-size: 13px;
}
button {
    width: 100%;
    margin-top:
    20px; padding:
    10px;
    background-color: #2563eb;
    color: white;
    border: none;
```



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```
#message {  
    text-align:  
        center;  
    font-weight: bold;  
    margin-top: 15px;  
}  
  
</style>  
</head>  
<body>  
<div class="form-box">  
    <h2>Shreya's student Registration Form</h2>  
    <form id="studentForm">  
        <label>Name:</label>  
        <input type="text" id="name">  
        <div id="nameError" class="error"></div>  
        <label>Email:</label>  
        <input type="text" id="email">  
        <div id="emailError" class="error"></div>  
        <label>Course:</label>  
        <select id="course">  
            <option value="">Select Course</option>  
            <option value="CSE">Computer  
Science</option>  
            <option value="IT">Information  
Technology</option>  
            <option value="ECE">Electronics</option>  
        </select>  
        <div id="courseError" class="error"></div>  
        <button type="submit">Submit</button>  
    </form>
```



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```
</div>

<script>

    // Access form elements
    let form =
document.getElementById("studentForm");

    let name = document.getElementById("name");
    let email = document.getElementById("email");
    let course = document.getElementById("course");
    // Focus event
    name.addEventListener("focus", function() {
        name.style.backgroundColor = "#eef4ff";
    });
    email.addEventListener("focus", function() {
        email.style.backgroundColor = "#eef4ff";
    });
    // Change event
    course.addEventListener("change", function() {
document.getElementById("courseError").innerText =
        "Selected Course: " + course.value;
    });

    // Submit event
    form.addEventListener("submit", function(event)
{
        event.preventDefault()
        ; let valid = true;
        // Name validation
        if (name.value.trim() === "") {
            document.getElementById("nameError").innerText
```



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```
        "Name cannot be empty";
        valid = false;
    } else {
document.getElementById("nameError").innerText =
"";
    }
    // Email validation
    let emailPattern =
/^^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (email.value.trim() === "") {
document.getElementById("emailError").innerText =
        "Email cannot be empty";
        valid = false;
    }
    else if (!emailPattern.test(email.value)) {
document.getElementById("emailError").innerText =
        "Enter a valid email";
        valid = false;
    }
    else {
document.getElementById("emailError").innerText =
"";
    }
    // Course validation
    if (course.value === "") {
```



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```
document.getElementById("courseError").innerText =
    "Please select a course";
    valid = false;
}
// Final result
if (valid) {
document.getElementById("message").innerText =
    "Registration submitted
successfully!";

document.getElementById("message").style.color =
"green";
}
else {

document.getElementById("message").innerText =
    "Please correct the errors.";
document.getElementById("message").style.color =
"red";
}
});
</script>
</body>
</html>
```



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3. Output:

Shreya's student Registration Form

Name:

Email:

Course:

Select Course

Submit

Shreya's student Registration Form

Name:

SHREYA MASKE

Email:

shreyamaske15@gmail.com

Course:

Computer Science

Selected Course: CSE

Submit

Registration submitted successfully!

4. Result/Conclusion:

Thus, JavaScript events were successfully handled using **focus**, **change**, and **submit**, and form elements were accessed, manipulated, and validated using JavaScript.



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● Experiment No.: 8

Title: Access and validate form fields using events; design a gym admission form with live input checks.

Software/Tools Required: VS Code, JavaScript.

Theory:

- i) The program creates a **Gym Admission Form** using HTML with fields for name, mobile number, email, age, and membership plan.
- ii) CSS is used to provide an attractive layout, colors, spacing, and styling to the form.
- iii) JavaScript uses **getElementById()** to access the form elements.
- iv) The **input event** performs live validation while the user enters data.
- v) The mobile number is restricted to **10 digits**, and **replace()** removes non-numeric characters.
- vi) A **Regular Expression** is used to validate the email format.
- vii) The age is validated using conditions to allow only values from 16 to 80.
- viii) The **change event** validates the selected membership plan.
- ix) The **submit event** validates all form details before registration.
preventDefault() prevents the form from being submitted or refreshing the page.
- x) if-else conditions determine whether the entered data is valid or invalid.
- xi) **innerHTML** displays **error and success messages** dynamically.
form.reset() clears all fields after successful registration.
- xii) Thus, the program demonstrates **DOM manipulation, event handling, live input validation, Regex, conditional statements, and form submission**.

Output :

Shreya's Gym Admission Form

Name:

Mobile Number:
✓ Valid mobile number

Email:
✓ Valid email

Age:
✓ Valid age

Membership Plan:
✓ Quarterly plan selected



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The screenshot shows a web browser window with the URL `File /Users/shreyamaske/javascript_exp/jsexp8/shreya_1196_prac8.html`. The form is titled "Shreya's Gym Admission Form" and contains the following fields and feedback:

- Name:**
- Mobile Number:**
Enter 10 digits
- Email:**
Enter a valid email
- Age:**
Age must be between 16 and 80
- Membership Plan:**
✓ Monthly plan selected

A blue "Submit" button is at the bottom. Below it, a red error message states: "✗ Please enter all details correctly."

The screenshot shows the same form after successful registration. The fields now contain placeholder text and show green validation feedback:

- Name:**
- Mobile Number:**
✓ Valid mobile number
- Email:**
✓ Valid email
- Age:**
✓ Valid age
- Membership Plan:**
✓ Quarterly plan selected

The blue "Submit" button remains. Below it, a green success message states: "✓ Registration Successful! Welcome, Shreya Maske!"

Result/Conclusion:

Thus, the gym admission form was successfully developed using **HTML, CSS, and JavaScript**. Form fields were accessed using the DOM, and **event handling with live validation** was implemented for mobile number, email, age, and membership plan. The form displays an appropriate success message only when all details are entered correctly.



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Case Study No.: 8

- Case Study Title:** Online Job Application Form, A company wants to create an online job application form that checks the entered information immediately and prevents submission when the details are invalid.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Job Application Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #dbeafe, #e0e7ff);
      margin: 0;
      padding: 25px;
    }
    .container {
      width: 650px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(0,0,0,0.15);
    }
    h2 {
      text-align: center;
      color: #3730a3;
    }
    h3 {
      color: #4f46e5;
      border-bottom: 2px solid #ddd;
      padding-bottom: 5px;
      margin-top: 25px;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Online Job Application Form</h2>
    <h3>Name</h3>
    <input type="text">
  </div>
</body>
</html>
```



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```
label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
}

input, select, textarea {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #ccc;
    border-radius: 6px;
    box-sizing: border-box;
}

textarea {
    height: 70px;
    resize: none;
}

.radio {
    width: auto;
    margin-right: 5px;
}

.message {
    font-size: 13px;
    margin-top: 4px;
}

.error {
    color: red;
}

.success {
    color: green;
}

button {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    background-color: #4f46e5;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    cursor: pointer;
}
```



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```
}

button:hover {

    background-color: #3730a3;

}

#result {

    text-align: center;

    margin-top: 20px;

    font-weight: bold;

}

</style>
</head>
<body>
<div class="container">

    <h2>📁 Shreya's 1196 Online Job Application Form</h2>

    <form id="jobForm">

        <h3>Personal Information</h3>

        <!-- NAME -->

        <label>Full Name:</label>

        <input type="text" id="name" placeholder="Enter full name">

        <div id="nameMessage" class="message"></div>

        <!-- DOB -->

        <label>Date of Birth:</label>

        <input type="date" id="dob">

        <div id="dobMessage" class="message"></div>

        <!-- GENDER -->

        <label>Gender:</label>

        <input type="radio" class="radio" name="gender" value="Male">
Male

        <input type="radio" class="radio" name="gender" value="Female">
Female

        <input type="radio" class="radio" name="gender" value="Other">
Other

        <div id="genderMessage" class="message"></div>

        <!-- EMAIL -->

        <label>Email Address:</label>

        <input type="text" id="email" placeholder="Enter email">

        <div id="emailMessage" class="message"></div>

        <!-- MOBILE -->

        <label>Mobile Number:</label>

        <input type="text" id="mobile"

            placeholder="Enter 10-digit mobile number"

    </div>

</form>

</div>

</body>

</html>
```



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```
maxlength="10">

<div id="mobileMessage" class="message"></div>

<!-- ADDRESS -->
<label>Address:</label>
<textarea id="address" placeholder="Enter address"></textarea>
<div id="addressMessage" class="message"></div>

<!-- CITY -->
<label>City:</label>
<input type="text" id="city" placeholder="Enter city">
<div id="cityMessage" class="message"></div>

<!-- STATE -->
<label>State:</label>
<input type="text" id="state" placeholder="Enter state">
<div id="stateMessage" class="message"></div>

<!-- PIN -->
<label>PIN Code:</label>
<input type="text" id="pin"
      placeholder="Enter 6-digit PIN"
      maxlength="6">
<div id="pinMessage" class="message"></div>

<!-- QUALIFICATION -->
<label>Educational Qualification:</label>
<select id="qualification">
  <option value="">Select qualification</option>
  <option>B.Tech</option>
  <option>B.E.</option>
  <option>BCA</option>
  <option>MCA</option>
  <option>M.Tech</option>
  <option>Other</option>
</select>
<div id="qualificationMessage" class="message"></div>

<!-- PERCENTAGE -->
<label>Percentage:</label>
<input type="number" id="percentage"
      placeholder="Enter percentage">
<div id="percentageMessage" class="message"></div>

<!-- PASSING YEAR -->
<label>Passing Year:</label>
<input type="number" id="passingYear"
      placeholder="Enter passing year">
```



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```
<div id="yearMessage" class="message"></div>

<h3>Professional Information</h3>

<!-- POSITION -->
<label>Applying for Job Position:</label>
<input type="text" id="position"
        placeholder="Example: Software Developer">
<div id="positionMessage" class="message"></div>

<!-- EXPERIENCE -->
<label>Total Work Experience (Years):</label>
<input type="number" id="experience"
        placeholder="Enter experience">
<div id="experienceMessage" class="message"></div>

<!-- ORGANIZATION -->
<label>Current / Previous Organization:</label>
<input type="text" id="organization"
        placeholder="Enter organization">
<div id="organizationMessage" class="message"></div>

<!-- DESIGNATION -->
<label>Current Designation:</label>
<input type="text" id="designation"
        placeholder="Example: Intern">
<div id="designationMessage" class="message"></div>

<!-- SALARY -->
<label>Expected Salary:</label>
<input type="number" id="salary"
        placeholder="Enter expected salary">
<div id="salaryMessage" class="message"></div>

<!-- SKILLS -->
<label>Technical Skills:</label>
<textarea id="skills"
        placeholder="Example: Java, Python, SQL,
HTML"></textarea>
<div id="skillsMessage" class="message"></div>
<button type="submit">Submit Application</button>
</form>

<div id="result"></div>
</div>

<script>

    let form = document.getElementById("jobForm");

    // FUNCTION TO SHOW MESSAGE

    function showMessage(id, message, type) {
```



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```
document.getElementById(id).className = "message " + type;
document.getElementById(id).innerHTML = message;

// NAME
document.getElementById("name").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("nameMessage", "Name is required", "error");
    } else {
        showMessage("nameMessage", "✓ Valid name", "success");
    }
});

// DATE OF BIRTH
document.getElementById("dob").addEventListener("change", function()
{
    if (this.value === "") {
        showMessage("dobMessage", "Date of birth is required",
"error");
    } else {
        showMessage("dobMessage", "✓ Date of birth selected",
"success");
    }
});

// GENDER
document.querySelectorAll('input[name="gender"]').forEach(function(radi
o) {
    radio.addEventListener("change", function() {
        showMessage("genderMessage", "✓ Gender selected",
"success");
    });
});

// EMAIL
document.getElementById("email").addEventListener("input",
function() {
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (pattern.test(this.value)) {
        showMessage("emailMessage", "✓ Valid email", "success");
    } else {
        showMessage("emailMessage", "Invalid email address",
"error");
    }
});
```



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```
    }

    });

    // MOBILE
    document.getElementById("mobile").addEventListener("input",
function() {
    this.value = this.value.replace(/[^0-9]/g, "");
    if (this.value.length === 10) {
        showMessage("mobileMessage", "✓ Valid mobile number",
"success");
    } else {
        showMessage("mobileMessage", "Invalid mobile number - enter
10 digits", "error");
    }
});

    // ADDRESS
    document.getElementById("address").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("addressMessage", "Address is required",
"error");
    } else {
        showMessage("addressMessage", "✓ Valid address",
"success");
    }
});

    // CITY
    document.getElementById("city").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("cityMessage", "City is required", "error");
    } else {
        showMessage("cityMessage", "✓ Valid city", "success");
    }
});

    // STATE
    document.getElementById("state").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("stateMessage", "State is required", "error");
    } else {
        showMessage("stateMessage", "✓ Valid state", "success");
    }
});
});
```



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```
    }

});

// PIN
document.getElementById("pin").addEventListener("input", function()
{
    this.value = this.value.replace(/^[^0-9]/g, "");
    if (this.value.length === 6) {
        showMessage("pinMessage", "✓ Valid PIN code", "success");
    } else {
        showMessage("pinMessage", "Invalid PIN - enter 6 digits",
"error");
    }
});

// QUALIFICATION
document.getElementById("qualification").addEventListener("change",
function() {
    if (this.value === "") {
        showMessage("qualificationMessage",
            "Please select qualification", "error");
    } else {
        showMessage("qualificationMessage",
            "✓ Qualification selected", "success");
    }
});

// PERCENTAGE
document.getElementById("percentage").addEventListener("input",
function() {
    let value = Number(this.value);
    if (value >= 0 && value <= 100) {
        showMessage("percentageMessage",
            "✓ Valid percentage", "success");
    } else {
        showMessage("percentageMessage",
            "Invalid percentage - enter 0 to 100", "error");
    }
});

// PASSING YEAR
document.getElementById("passingYear").addEventListener("input",
function() {
    let year = Number(this.value);
    let currentYear = new Date().getFullYear();
```



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```
if (year >= 1950 && year <= currentYear) {
    showMessage("yearMessage",
        "✓ Valid passing year", "success");
} else {
    showMessage("yearMessage",
        "Invalid passing year", "error");
}
});
// JOB POSITION
document.getElementById("position").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("positionMessage",
            "Job position is required", "error");
    } else {
        showMessage("positionMessage",
            "✓ Valid job position", "success");
    }
});
// EXPERIENCE
document.getElementById("experience").addEventListener("input",
function() {
    let value = Number(this.value);

    if (value >= 0) {
        showMessage("experienceMessage",
            "✓ Valid experience", "success");
    } else {
        showMessage("experienceMessage",
            "Experience cannot be negative", "error");
    }
});
// ORGANIZATION
document.getElementById("organization").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("organizationMessage",
            "Organization is required", "error");
    } else {
        showMessage("organizationMessage",
```



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```
        "✓ Valid organization", "success");

    }

});

// DESIGNATION
document.getElementById("designation").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("designationMessage",
            "Designation is required", "error");
    } else {
        showMessage("designationMessage",
            "✓ Valid designation", "success");
    }
});

// SALARY
document.getElementById("salary").addEventListener("input",
function() {
    if (Number(this.value) > 0) {
        showMessage("salaryMessage",
            "✓ Valid expected salary", "success");
    } else {
        showMessage("salaryMessage",
            "Enter a valid salary", "error");
    }
});

// TECHNICAL SKILLS
document.getElementById("skills").addEventListener("input",
function() {

    if (this.value.trim() === "") {
        showMessage("skillsMessage",
            "Technical skills are required", "error");
    } else {
        showMessage("skillsMessage",
            "✓ Skills entered", "success");
    }
});
```



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```
// FORM SUBMISSION

form.addEventListener("submit", function(event) {

    event.preventDefault();

    let name = document.getElementById("name").value.trim();
    let dob = document.getElementById("dob").value;
    let gender =
document.querySelector('input[name="gender"]:checked');
    let email = document.getElementById("email").value;
    let mobile = document.getElementById("mobile").value;
    let address = document.getElementById("address").value.trim();
    let city = document.getElementById("city").value.trim();
    let state = document.getElementById("state").value.trim();
    let pin = document.getElementById("pin").value;
    let qualification =
document.getElementById("qualification").value;
    let percentage =
Number(document.getElementById("percentage").value);
    let passingYear =
Number(document.getElementById("passingYear").value);

    let position = document.getElementById("position").value.trim();
    let experience =
Number(document.getElementById("experience").value);
    let organization =
document.getElementById("organization").value.trim();
    let designation =
document.getElementById("designation").value.trim();
    let salary = Number(document.getElementById("salary").value);
    let skills = document.getElementById("skills").value.trim();

    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    let currentYear = new Date().getFullYear();
    if (
        name !== "" &&
        dob !== "" &&
        gender &&
        emailPattern.test(email) &&
        mobile.length === 10 &&
        address !== "" &&
        city !== "" &&
        state !== "" &&
```



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```
pin.length === 6 &&
qualification !== "" &&
percentage >= 0 &&
percentage <= 100 &&
passingYear >= 1950 &&
passingYear <= currentYear &&
position !== "" &&
experience >= 0 &&
organization !== "" &&
designation !== "" &&
salary > 0 &&
skills !== ""
) {
    document.getElementById("result").className = "success";

    document.getElementById("result").innerHTML =
        "✔ Application Submitted Successfully!";

    form.reset();

} else {

    document.getElementById("result").className = "error";
    document.getElementById("result").innerHTML =
        "✘ Please correct the invalid details before
submitting.";
    }
});
</script>
</body>
</html>
```

3. Output:



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File /Users/shreyamaske/javascript_exp/jsexp8/shreya_1196_case8.html



Shreya's 1196 Online Job Application Form

Personal Information

Full Name:

Shreya Maske

✓ Valid name

Date of Birth:

15 / 06 / 2006

✓ Date of birth selected

Gender:

☐ Male ☒ Female ☐ Other

✓ Gender selected

Email Address:

shreya@gmailcom

Invalid email address

Mobile Number:

7765211679

✓ Valid mobile number

Address:

412, Baker's Street

✓ Valid address

File /Users/shreyamaske/javascript_exp/jsexp8/shreya_1196_case8.html



City:

Nagpur

✓ Valid city

State:

Maharashtra

✓ Valid state

PIN Code:

44006

Invalid PIN - enter 6 digits

Educational Qualification:

B.Tech

✓ Qualification selected

Percentage:

88

✓ Valid percentage

Passing Year:

2028

Invalid passing year



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← → ↻ ⓘ File /Users/shreyamaske/javascript_exp/jsxp8/shreya_1196_case8.html

Professional Information

Applying for Job Position:

Software Developer

✓ Valid job position

Total Work Experience (Years):

-1

Experience cannot be negative

Current / Previous Organization:

xyz

✓ Valid organization

Current Designation:

Intern

✓ Valid designation

Expected Salary:

800000

✓ Valid expected salary

Technical Skills:

Java

✓ Skills entered

Submit Application

✗ Please correct the invalid details before submitting.

4. Result/Conclusion:

Thus, an online job application form was successfully created with **live validation** to check user details and prevent invalid form submission.



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● Experiment No.: 9

1. **Title:** Implement localStorage and sessionStorage to save user preferences like theme settings.

2. **Software/Tools Required:** VS Code, JavaScript,html.

3. Theory:

- 1) Web Storage allows websites to store data in the browser using key-value pairs.
- 2) localStorage stores data persistently and remains available even after closing the browser.
- 3) sessionStorage stores data only for the current browser session.
- 4) setItem() is used to save the selected theme and storage type.
- 5) getItem() retrieves previously saved theme and storage information.
- 6) removeItem() removes stored preferences.
- 7) The program provides Light, Dark, and Blue themes using CSS classes.
- 8) classList.remove() and classList.add() are used to change the theme dynamically.
- 9) getElementById() accesses the required HTML elements.
- 10) innerText displays the current theme, storage type, saved value, and messages.
- 11) loadPreference() automatically retrieves saved preferences when the page loads.
- 12) savePreference() stores the selected theme in either localStorage or sessionStorage.
- 13) removePreference() clears the saved preferences and resets the theme to Light.
- 14) The program demonstrates web storage and dynamic DOM manipulation.



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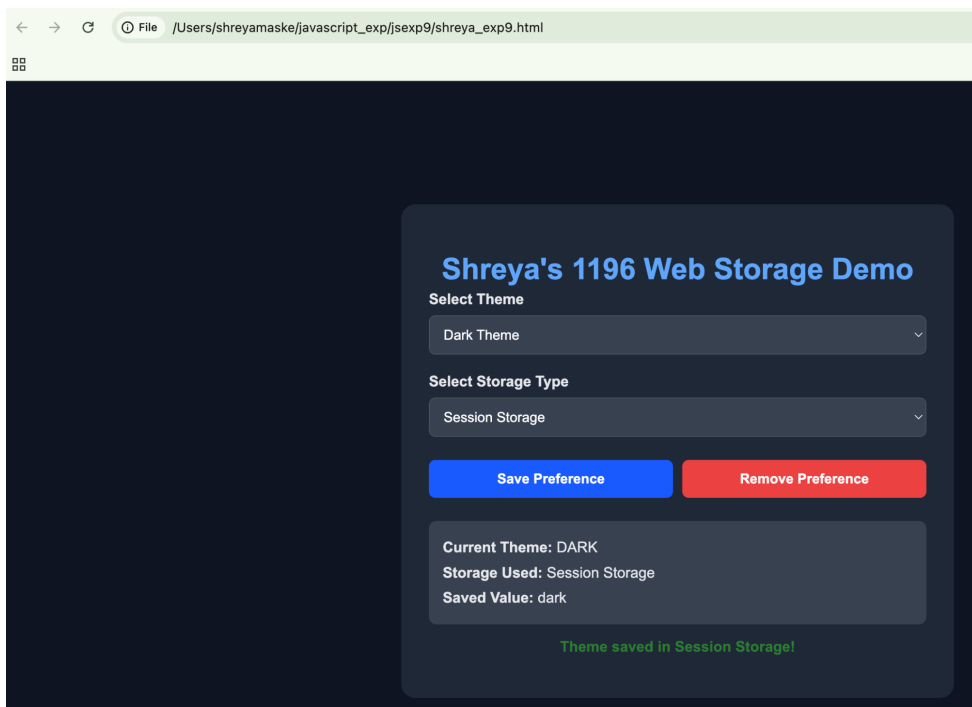
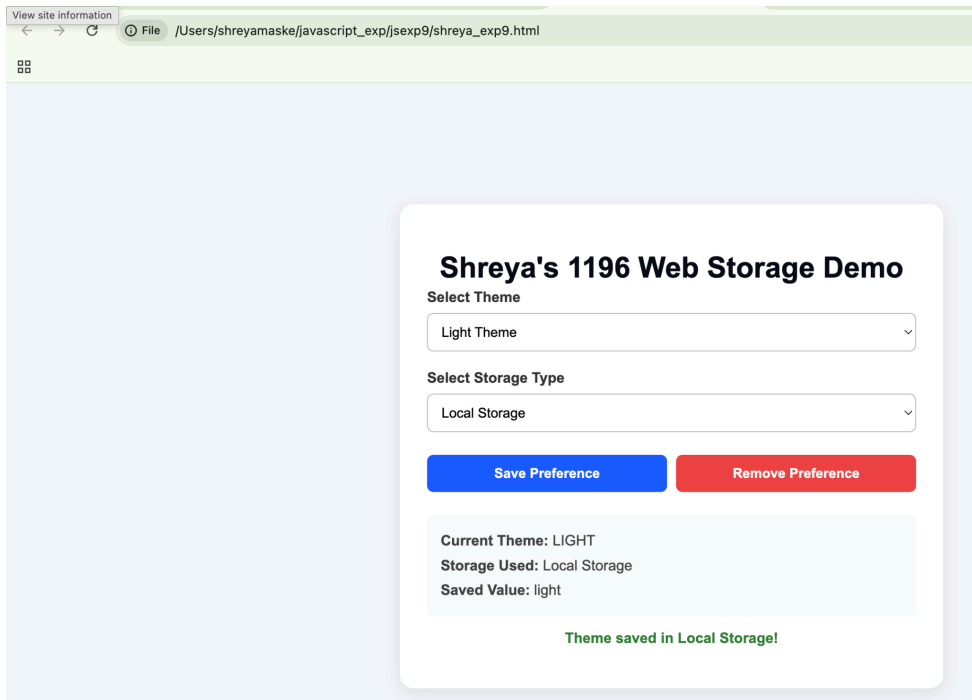
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4. Output :



5. Result/Conclusion:

Thus, local Storage and session Storage were successfully implemented to store and retrieve user theme preferences. The selected theme was applied dynamically, displayed on the webpage, automatically loaded when available, and removed successfully using JavaScript.



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● Experiment No.: 10

1. **Title:** Load and display JSON data using fetch() and jQuery \$.getJSON() ; render results dynamically in a table.

2. **Software/Tools Required:** VS Code, JavaScript,html.

3. Theory:

- 1) String functions in JavaScript are used to manipulate and analyze text.
- 2) The student data is stored as JSON objects containing id, name, and prn.
- 3) fetch() is a JavaScript method used to request and retrieve data asynchronously.
- 4) response.json() converts the fetched response into JavaScript data.
- 5) jQuery \$.getJSON() is used to load JSON data easily using jQuery.
- 6) forEach() is used to process each student object from the JSON data.
- 7) createElement() creates table rows dynamically.
- 8) innerHTML inserts the student ID, name, and PRN into table cells.
- 9) appendChild() adds the dynamically created rows to the table.
- 10) Thus, the table is updated dynamically without manually writing each student record in HTML.



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4. Output :

The screenshots show a web application interface with a title 'Shreya's JSON Data' and two buttons: 'Load using fetch()' and 'Load using \$.getJSON()'. Below the buttons is a table with three columns: ID, Name, and PRN. The table contains three rows of student data.

ID	Name	PRN
1	Shreya	24070521196
2	Sharvari	24070521207
3	Vithika	24070521192

5. Result/Conclusion:

Thus, JSON student data containing ID, Name, and PRN was successfully loaded using both `fetch()` and `jQuery $.getJSON()` and dynamically displayed in an HTML table.